Unit 21

Oligopoly

After completing this unit, the student will be able to:

-Understand the features of Oligopoly

-Analyse the Interdependency of firms in price and output determination under oligopoly

Oligopoly

The term oligopoly is derived from two Greek words "oligos" meaning few and "pollen" meaning to sell. Oligopoly is that form of imperfect competition where there are a few firms in the market producing either homogenous or differentiated products. In other words, an oligopoly is a market structure in which a few firms dominate. When a market is shared between a few firms, it is said to be highly concentrated. Modern economists used the term concentration ratio to define monopoly. It represents the combined market share of the largest four firms in the market. Although only a few firms dominate, it is possible that many small firms may also operate in the market.

Classification of oligopoly

Oligopolies can be categorized into several types, primarily based on product differentiation and competitive behaviour.

Table 21.1

Basis of classification	Types of oligopoly	Basic feature
Nature of product	Perfect oligopoly	Selling Homogenous product
	Imperfect Differentiated oligopoly)	Selling Differentiated products
Freedom to entry and Exit	Open oligopoly	Existence of freedom to enter in the market
	Closed oligopoly	No freedom to entry and exit

On the basis of Agreement or understanding	Collusive oligopoly	Agreement or understanding between firms (Explicit and Tacit)
	Non-collusive oligopoly	Lack of understanding or agreement
Price leadership	Partial Oligopoly	One large firm (Price leadership)
	Full oligopoly	No firm is a price leader
Degree of Co ordination	Syndicated oligopoly	Sell the products of firms through a centralised syndicate
	Organised oligopoly	Organise themselves into a central association for fixing output price etc
Concentration Ratio	Tight oligopoly	Concentration ratio 60 %
	Loose oligopoly	Concentration ratio 40% to 20%

Important features of oligopoly

The main characteristics of firms operating in a market with few close rivals include:

1. Interdependence:

Firms that are interdependent cannot act independently of each other. A firm operating in a market with just a few competitors must take the potential reaction of its closest rivals into account when making its own decisions. For example, if a petrol retailer like Indian Oil Corporation wishes to increase its market share by reducing price, it must take into account the possibility that close rivals, such as HP and BP, may reduce their price in retaliation. An understanding of game theory and the Prisoner's Dilemma helps appreciate the concept of interdependence.

2. Strategy

Strategy is extremely important to firms that are interdependent. Because firms cannot act independently, they must anticipate the likely response of a rival to any given change in their price, or their non-price activity. In other words, they need to plan, and work out a range of

possible options based on how they think rivals might react. Oligopolists have to make critical strategic decisions, such as:

- Whether to compete with rivals, or collude with them.
- Whether to raise or lower price, or keep price constant.
- Whether to be the first firm to implement a new strategy, or whether to wait and see what rivals do.

The advantages of 'going first' or 'going second' are respectively called 1st and 2nd -mover advantage. Sometimes it pays to go first because a firm can generate head-start profits. 2nd mover advantage occurs when it pays to wait and see what new strategies are launched by rivals, and then try to improve on them or find ways to undermine them.

3. Barriers to entry

Oligopolies and monopolies frequently maintain their position of dominance in a market might because it is too costly or difficult for potential rivals to enter the market. These hurdles are called barriers to entry and the incumbent can erect them deliberately, or they can exploit natural barriers that exist.

1) Natural entry barriers include:

a. Economies of large-scale production:

If a market has significant economies of scale that have already been exploited by the incumbents, new entrants are deterred.

b. Ownership or control of a key scarce resource**:**

Owning scarce resources that other firms would like to use creates a considerable barrier to entry, such as an airline controlling access to an airport.

c. High set-up costs:

High set-up costs deter initial market entry, because they increase break-even output, and delay the possibility of making profits. Many of these costs are sunk costs, which are costs that cannot be recovered when a firm leaves a market, and include marketing and advertising costs and other fixed costs.

d. High R&D costs:

Spending money on Research and Development (R & D) is often a signal to potential entrants that the firm has large financial reserves. In order to compete, new entrants will have to match, or exceed, this level of spending in order to compete in the future. This deters entry, and is widely found in oligopolistic markets such as pharmaceuticals and the chemical industry.

2) Artificial barriers include:

a. Predatory pricing:

Predatory pricing occurs when a firm deliberately tries to push prices low enough to force rivals out of the market.

b. Limit pricing:

Limit pricing means the incumbent firm sets a low price, and a high output, so that entrants cannot make a profit at that price. This is best achieved by selling at a price just below the average total costs (ATC) of potential entrants. This signals to potential entrants that profits are impossible to make.

c. Superior knowledge:

An incumbent may, over time, have built up a superior level of knowledge of the market, its customers, and its production costs. This superior knowledge can deter entrants into the market.

d. Predatory acquisition:

Predatory acquisition involves taking-over a potential rival by purchasing sufficient shares to gain a controlling interest, or by a complete buy-out. As with other deliberate barriers, regulators, like the Competition Commission, may prevent this because it is likely to reduce competition.

e. Advertising:

Advertising is another sunk cost - the more that is spent by incumbent firms the greater the deterrent to new entrants.

f. A strong brand:

A strong brand creates loyalty, 'locks in' existing customers, and deters entry.

g. Loyalty schemes:

Schemes such as Tesco's Club Card, help oligopolists retain customer loyalty and deter entrants who need to gain market share.

h. Exclusive contracts, patents and licences:

These make entry difficult as they favour existing firms who have won the contracts or own the licenses. For example, contracts between suppliers and retailers can exclude other retailers from entering the market.

i. Vertical integration:

Vertical integration can 'tie up' the supply chain and make life tough for potential entrants, such as an electronics manufacturer like Sony having its own retail outlets (Sony Centres

4. Indeterminateness of demand curve facing an oligopolist:

The interdependence between firms results in the indeterminateness of AR and MR curves faced by an oligopolistic market.

5. Conflicting attitudes of the firms

6. Element of monopoly

7. Indeterminateness of demand curve:

Due to strong interdependence between firms, the demand curve becomes indeterminate. The uncertainty prevailing in the market restricts a firm to follow an independent price policy and thus demand curve cannot be determined.

8. Price rigidity:

The basic feature of oligopoly market is that their price is rigid. If a firm tend to decrease the price of their product, all other firms will follow it, and no firm will be benefited with the price decrease. If a firm increases the price of their product, no other firm will follow it and the price increased firm will lose their customers. So, a firm under oligopoly has no tendency either to increase or decrease their price.

Collusive versus non-collusive oligopoly

Another key feature of oligopolistic markets is that firms may attempt to collude, rather than compete. If colluding, participants act like a monopoly and can enjoy the benefits of higher profits over the long term.

Types of collusion

Overt: Overt collusion occurs when there is no attempt to hide agreements, such as the when firms form trade associations like the Association of Petrol Retailers.

Covert: Covert collusion occurs when firms try to hide the results of their collusion, usually to avoid detection by regulators, such as when fixing prices.

Tacit: Tacit collusion arises when firms act together, called acting in concert, but where there is no formal or even informal agreement. For example, it may be accepted that a particular firm is the price leader in an industry, and other firms simply follow the lead of this firm. All firms may ‘understand’ this, but no agreement or record exists to prove it. If firms do collude, and their behaviour can be proven to result in reduced competition, they are likely to be subject to

regulation. In many cases, tacit collusion is difficult or impossible to prove, though regulators are becoming increasingly sophisticated in developing new methods of detection.

Non-Collusive or Competitive oligopolies

When competing, oligopolists prefer non-price competition in order to avoid price wars. A price reduction may achieve strategic benefits, such as gaining market share, or deterring entry, but the danger is that rivals will simply reduce their prices in response. In such a case no conditions of collusion exist. This leads to little or no gain, but can lead to falling revenues and profits. Hence, a far more beneficial strategy may be to undertake non-price competition.

Cartels

Cartel is a collusive oligopoly model. Oligopolistic firms have an incentive to form cartels in which they collude in setting prices or quantities so as to increase their profits. The Organization of Petroleum Exporting Countries (OPEC) is a well-known example of an international cartel; however, many cartels operate within a single country. Typically, each member of a cartel agrees to produce less output than it would if it acted independently. As a result, the market price rises and the firms earn higher profits. If the firms reduce market output to the monopoly level, they achieve the highest possible collective profit. Luckily for consumers, cartels often fail because of government policies that forbid cartels or because members of the cartel "cheat" on the agreement. Each member has an incentive to cheat because it can raise its profit if it increases its output while other cartel members stick to the agreement.

A cartel forms if members of the cartel believe that they can raise their profits by coordinating their actions. When a firm acts independently, it considers how increasing its output affects its own profit only. The firm does not care that when it increases its output, it lowers the profits of other firms. A cartel, in contrast, considers how changes in any one firm's output affect the profits of all members of the cartel. Therefore, the aggregate profit of a cartel can exceed the combined profits of the same firms acting independently.

Although cartels are most common in oligopolistic markets, occasionally we see cartels formed in what would otherwise be highly competitive markets, such as in markets of professionals. If a competitive firm lowers its output, it raises the market price very slightly—so slightly that the firm ignores the effect not only on other firms' profits but also on its own. If all the identical competitive firms in an industry lower their output by this same amount, however, the market

price will change noticeably. Recognizing this effect of collective action, a cartel chooses to produce a smaller market output than is produced by a competitive market.

Figure 21.1 illustrates this difference between a competitive market and a cartel. This oligopolistic market has n firms, and no further entry is possible. Panel (a) show the marginal and average cost curves of a typical perfectly competitive firm. If all firms are price takers, the market supply curve, S in panel (b), is the horizontal sum of the individual marginal cost curves above minimum average cost. At the competitive price, p_c, each price-taking firm produces q_c units of output (which is determined by the intersection in panel (a) of MC and the dotted line at p_c. The market output is $Q_c = nq_c$ (where S intersects the market demand curve in panel b).

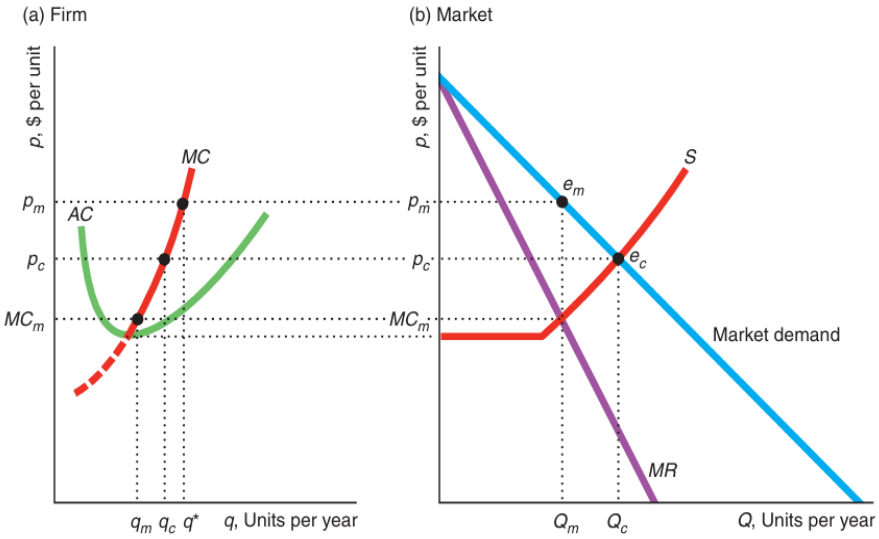


Figure 21.1: Competition Versus Cartel

Now suppose that the firms form a cartel. At the competitive output, the cartel's marginal cost (which is the competitive industry supply curve, S in panel (b)) is greater than its marginal revenue, so the cartel's profit rises if it reduces output. The cartel's collective profit rises until output is reduced by enough that its marginal revenue equals its marginal cost at Q_m, the monopoly output. If the profit of the cartel increases, the profit of each of the n members of the cartel also increases. To achieve the cartel output level, each firm must reduce its output to $q_m = Q_m/n$, as panel (a) shows.

A competitive firm produces q_c, where its marginal cost equals the market price. If only one firm reduces its output, it loses profit because it sells fewer units at essentially the same price. By getting all the firms to lower their output together, the cartel raises the market price and hence individual firms' profits. The less elastic the market demand the potential cartel faces,

all else the same, the higher the price the cartel sets and the greater the benefit from cartelizing. If the penalty for forming an illegal cartel is relatively low, some unscrupulous businesspeople may succumb to the lure of extra profits and join. So, the firms form a cartel to achieve these higher profits.

Laws Against Cartels

In the late nineteenth century, cartels (or trusts) were legal and common in the United States. In response to the trusts' high prices, the U.S. Congress passed the Sherman Antitrust Act in 1890 and the Federal Trade Commission Act of 1914, which prohibit firms from explicitly agreeing to take actions that reduce competition. In particular, cartels that are formed for the purpose of jointly setting price are strictly prohibited. By imposing penalties on firms caught colluding, government agencies seek to discourage cartels from forming. Cartels persist despite these laws for three reasons. **First**, international cartels and cartels within certain countries operate legally. Some international cartels organized by countries rather than firms operate legally. An example is the OPEC cartel, which was formed in 1960 by five major oil-exporting countries: Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela.

Second, some illegal cartels operate believing that they can avoid detection or, if caught, that the punishment will be insignificant. At least until recently, they were often correct. **Third**, some firms are able to coordinate their activities without explicitly colluding and thereby running afoul of competition laws. To determine guilt, U.S. antitrust laws use evidence of conspiracy—explicit agreements—rather than the economic effect of the suspected cartel. Charging monopoly-level prices is not necessarily illegal—only the "bad behaviour" of explicitly agreeing to raise prices is against the law. As a result, some groups of firms charge monopoly-level prices without violating competition laws. These firms may tacitly collude without meeting by signalling to each other through their actions. If one firm raises its price and keeps it high only if other firms follow its lead, it is not necessarily violating the law if the firms did not explicitly communicate.

However, in recent years, most developed countries have followed Canada and the United States in strictly prohibiting cartels. The European Union's competition policy under the Treaty of the European Community (EC Treaty or Treaty of Rome) in 1957 gives the European Union and member states substantial civil powers to prevent actions that hinder competition, including formation of cartels. Again, price-fixing is per se illegal. Increasingly, antitrust authorities from

around the world are cooperating. Cooperation agreements exist between authorities in Canada, Mexico, Europe, Australia, New Zealand, and the United States, among others. Such cooperation is critical given the increasingly global scope of the firms engaged in collusion and other anticompetitive activities.

Why Cartels Fail

Many cartels fail even without legal intervention. Cartels fail if noncartel members can supply consumers with large quantities of goods. For example, copper producers formed an international cartel that controlled only about a third of the world's copper production and faced additional competition from firms that recycle copper from scrap materials. Because of this competition from noncartel members, the cartel was not successful in setting and maintaining a cartel price. In addition, each member of a cartel has an incentive to cheat on the cartel agreement.

Figure 21.1 illustrates why firms want to cheat. At the cartel output, q_m in panel a, each cartel member's marginal cost is MC_m. The marginal revenue of a firm that violates the agreement is p_m because it is acting like a price taker with respect to the market price. Because the firm's marginal revenue (price) is above its marginal cost, the firm wants to increase its output. If the firm decides to violate the cartel agreement, it maximizes its profit by increasing its output to q*, where its marginal cost equals p_m. As more and more firms leave the cartel, the cartel price falls. The colluding firms act like a dominant firm facing a competitive fringe. Eventually, if enough firms quit, the cartel collapses.

Maintaining Cartels

To keep firms from violating the cartel agreement, the cartel must be able to detect cheating and punish violators. Further, the members of the cartel must keep their illegal behaviour hidden from customers and government agencies.

a) Detection and Enforcement

Cartels use various techniques to detect cheating. Some cartels, for example, give members the right to inspect each other's accounts. Cartels may also divide the market by region or by customers, making it more likely that a firm that steals another firm's customers is detected.

Another option is for a cartel to turn to industry organizations that collect data on market share by firm. A cheating cartel's market share would rise, tipping off the other firms that it cheated. Various methods are used to enforce cartel agreements.

For example, GE and Westinghouse, the two major sellers of large steam-turbine generators, included "most-favoured-customer" clauses in their contracts. These contracts stated that the seller would not offer a lower price to any other current or future buyer without offering the same price decrease to the firms that signed these contracts. This type of rebate clause creates a penalty for cheating on the cartel. Threats of violence are another means of enforcing a cartel agreement.

b) Government Support

Sometimes governments help create and enforce cartels, exempting them from antitrust and competition laws. The international airline market provides an example where governments first created a cartel and then later acted to end it. In 1944, 52 countries signed the Convention on International Civil Aviation, which established rules ("freedoms") that enabled airlines to fly between countries. International airfares were negotiated through bilateral governmental agreements rather than determined by the market, and airlines were given exemption from cartel laws, which allowed them to discuss prices through the International Air Transport Association (IATA). In the late 1970s, the United States deregulated its airline industry. Soon thereafter, European countries started to deregulate, allowing nongovernment-owned airlines to enter the market. Countries negotiated bilateral open skies agreements that weakened IATA's price fixing role.

c) Barriers to Entry

Barriers to entry that limit the number of firms help the cartel detect and punish cheating and keep prices high. The fewer the firms in a market, the more likely it is that other firms will know if a given firm cheats and the easier it is to impose costs on that firm. Cartels with a large number of firms are relatively rare, except those involving professional associations.

Noncooperative Oligopoly

A non-cooperative oligopoly is a market structure characterized by a small number of large firms that compete with each other rather than collude. This means each firm makes

independent decisions about pricing and output, considering the potential reactions of its rivals but not explicitly coordinating actions with them.

Key characteristics

- Few large firms: A handful of dominant firms control a significant portion of the market.
- Interdependence: Firms are highly interdependent; the decisions of one firm significantly influence the choices and outcomes of other firms.
- Competition: Despite interdependence, firms strive to maximize their individual profits and compete vigorously through various strategies such as pricing, advertising, and product differentiation.
- No formal agreements: Firms do not engage in formal collusion or agreements to fix prices, output, or market shares.
- Barriers to entry: High barriers to entry, such as large capital requirements, economies of scale, or brand loyalty, make it difficult for new firms to enter the market and compete with existing players.

Types of non-cooperative models

Several models attempt to explain the behaviour of firms in a non-cooperative oligopoly:

a) Cournot Model: Firms compete by choosing their output levels simultaneously and independently, assuming the output of their rivals is fixed. The equilibrium results in output and prices lower than a monopoly but higher than perfect competition.

b) Bertrand Model: Firms compete on price, assuming their rivals' prices will remain unchanged. In theory, this can lead to a perfectly competitive outcome where prices equal marginal costs and firms earn zero economic profits.

c) Stackelberg Model: One firm acts as a leader and sets its output level first, and other firms (followers) react by adjusting their output accordingly. The leader firm typically gains a strategic advantage and earns higher profits.

d) Kinked Demand Curve Model (Sweezy Model): This model attempts to explain price rigidity in oligopolistic markets. It suggests firms believe that if they raise prices,

competitors will not follow, leading to a loss of market share. However, if they lower prices, competitors will match the decrease, resulting in reduced profits for all. This creates a stable price at the 'kink' point.

e) Contestable Market Model: This model emphasizes the threat of potential entry and exit from the market, even with a few firms. If entry and exit barriers are low, existing firms may behave more competitively to deter new entrants.

Comparison with cooperative oligopoly

The key difference with cooperative oligopoly lies in firms' willingness to cooperate. That means in Non-cooperative oligopoly firms compete independently, each aiming to maximize its own profit. But in Cooperative oligopoly firms cooperate, formally or informally, to reduce competition and maximize their joint profit. This often involves forming cartels to set prices and output quotas. Many industries operate as non-cooperative oligopolies. For example, in automobile industry companies like Toyota, Volkswagen, and General Motors compete fiercely through product differentiation, advertising, and innovation. Whereas in the smartphone market, dominated firms like Apple, Samsung, and a few others, who engage in non-price competition and strategic pricing. But in the airline industry, a handful of major airlines compete for passengers on routes, often leading to price sensitivity and strategic reactions to competitor pricing. However, in mass media, a small number of media conglomerates control a large share of the market, impacting news and entertainment content.

Cournot Oligopoly

A simple model of duopoly first introduced by the French economist Augustin Cournot in 1838. Suppose the firms produce a homogeneous good and know the market demand curve. Each firm must decide how much to produce, and the two firms make their decisions at the same time. When making its production decision, each firm takes its competitor into account. It knows that its competitor is also deciding how much to produce, and the market price will depend on the total output of both firms. The essence of the Cournot model is that each firm treats the output level of its competitor as fixed when deciding how much to produce. To see how this works, let's consider the output decision of Firm 1. Suppose Firm 1 thinks that Firm 2 will produce nothing. In that case, Firm 1's demand curve is the market demand curve. In

Figure 21.2 this is shown as $D_1(0)$, which means the demand curve for Firm 1, assuming Firm 2 produces zero. It also shows the corresponding marginal revenue curve $MR_1(0)$. We have assumed that Firm 1's marginal cost MC_1 is constant. As shown in the figure 21.2, Firm 1's profit-maximizing output is 50 units, the point where $MR_1(0)$ intersects MC_1. So, if Firm 2 produces zero, Firm 1 should produce 50.

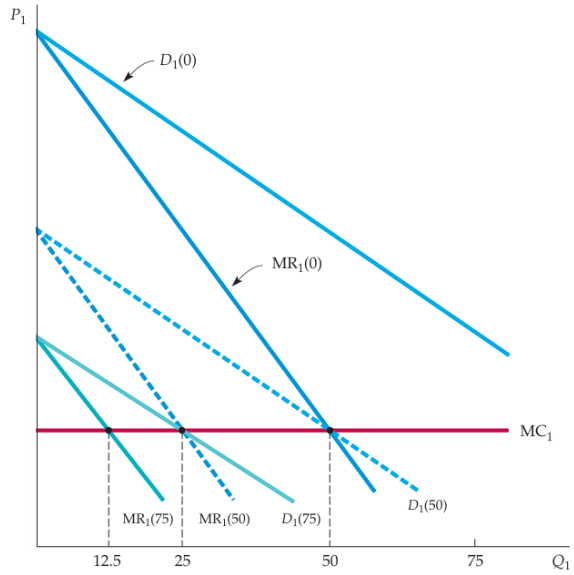


Figure 21.2

Suppose, instead, that Firm 1 thinks Firm 2 will produce 50 units. Then Firm 1's demand curve is the market demand curve shifted to the left by 50. In Figure 21.2, this curve is labelled $D_1(50)$, and the corresponding marginal revenue curve is labelled $MR_1(50)$. Firm 1's profit-maximizing output is now 25 units, the point where $MR_1(50) = MC_1$. Now, suppose Firm 1 thinks that Firm 2 will produce 75 units. Then Firm 1's demand curve is the market demand curve shifted to the left by 75. It is labelled $D_1(75)$ in Figure 21.2, and the corresponding marginal revenue curve is labelled $MR_1(75)$. Firm 1's profit-maximizing output is now 12.5 units, the point where $MR_1(75) = MC_1$. Finally, suppose Firm 1 thinks that Firm 2 will produce 100 units. Then Firm 1's demand and marginal revenue curves (which are not shown in the figure) would intersect its marginal cost curve on the vertical axis; if Firm 1 thinks that Firm 2 will produce 100 units or more, it should produce nothing.

Reaction Curves

If Firm 1 thinks that Firm 2 will produce nothing, it will produce 50; if it thinks Firm 2 will produce 50, it will produce 25; if it thinks Firm 2 will produce 75, it will produce 12.5; and if it thinks Firm 2 will produce 100, then it will produce nothing. Firm 1's profit-maximizing output is thus a decreasing schedule of how much it thinks Firm 2 will produce. We call this schedule Firm 1's reaction curve and denote it by $Q_1^*(Q_2)$. This curve is plotted in Figure 21.3, where each of the four output combinations that we found above is shown as an x.

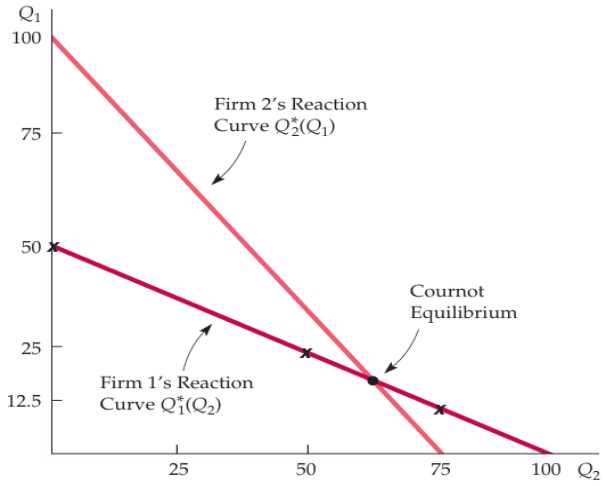


Figure 21.3

We can go through the same kind of analysis for Firm 2; that is, we can determine Firm 2's profit-maximizing quantity given various assumptions about how much Firm 1 will produce. The result will be a reaction curve for Firm 2-i.e., a schedule $Q_2^*(Q_1)$ that relates its output to the output that it thinks Firm 1 will produce. If Firm 2's marginal revenue or marginal cost curve is different from that of Firm 1, its reaction curve will also differ in form. For example, Firm 2's reaction curve might look like the one drawn in Figure 21.3.

The Cournot Equilibrium and the Number of Firms

Equilibrium in the Cournot model in which each firm correctly assumes how much its competitor will produce and sets its own production level accordingly. Each firm's reaction curve tells it how much to produce, given the output of its competitor. In equilibrium, each firm sets output according to its own reaction curve; the equilibrium output levels are therefore

found at the intersection of the two reaction curves. We call the resulting set of output levels a Cournot equilibrium. In this equilibrium, each firm correctly assumes how much its competitor will produce, and it maximizes its profit accordingly.

This Cournot equilibrium is an example of a Nash equilibrium (and thus it is sometimes called a Cournot-Nash equilibrium). In a Nash equilibrium, each firm is doing the best it can give what its competitors are doing. As a result, no firm would individually want to change its behaviour. In the Cournot equilibrium, each firm is producing an amount that maximizes its profit given what its competitor is producing, so neither would want to change its output.

Suppose the two firms are initially producing output levels that differ from the Cournot equilibrium. Unfortunately, the Cournot model says nothing about the dynamics of the adjustment process. In fact, during any adjustment process, the model's central assumption that each firm can assume that its competitor's output is fixed will not hold. Because both firms would be adjusting their out puts, neither output would be fixed. It is rational if the two firms are choosing their outputs only once because then their outputs cannot change. It is also rational once they are in Cournot equilibrium because then neither firm will have any incentive to change its output.

In a Cournot model of oligopoly, as the number of firms competing in the market increases, the total output produced by the market tends to increase, while the market price and individual firm profits tend to decrease. When the number of firms approaches infinity, the Cournot equilibrium converges towards a perfectly competitive market outcome.

More firms in the market lead to greater competition, pushing firms to produce more output to capture market share. But increased supply due to more firms results in a lower market price. Lower prices translate to lower individual firm profits, as they have less control over pricing. As the number of firms becomes very large, the market essentially mimics a perfectly competitive market, with firms producing at a level where price equals marginal cost and economic profits are driven to zero.

Example:

Consider a Cournot duopoly (two firms). Each firm will choose its output level to maximize its profits, taking the other firm's output as given. As the number of firms increases, each firm will face more competitors, leading to lower prices and profits for all. In essence, the Cournot model demonstrates a spectrum of market outcomes, from near-monopoly with a few firms to

near-perfect competition with many firms, with the key factor being the level of competition among the firms.

Stackelberg Graphical Model

In the Cournot model, both firms make their output decisions at the same time. Suppose, however, that one of the firms, called the leader, can set its output before its rival, the follower, sets its output. Having one firm act before another arises naturally if one firm enters the market first. Examples of leaders include IBM in the mainframe computer market, General Electric in the turbine generator industry, and General Foods among coffee roasters. The leader realizes that once it sets its output, the rival firm will use its Cournot best-response curve to pick a best-response output. Thus, the leader predicts what the follower will do before the follower acts. Using this knowledge, the leader manipulates the follower, thereby benefiting at the follower's expense. This model was developed by the German economist Heinrich von Stackelberg and is an extension of Cournot's model. It is assumed, by von Stackelberg, that one duopolist is sufficiently sophisticated to recognise that his competitor acts on the Cournot assumption. This recognition allows the sophisticated duopolist to determine the reaction curve of his rival and incorporate it in his own profit function, which he then proceeds to maximise like a monopolist.

Assume that the isoprofit curves and the reaction functions of the duopolists are those depicted in figure 21.4. If firm A is the sophisticated oligopolist, it will assume that its rival will act on the basis of its own reaction curve. This recognition will permit firm A to choose to set its own output at the level which maximises its own profit. This is point A (in figure 21.4) which lies on the lowest possible isoprofit curve of A, denoting the maximum profit A can achieve given B's reaction curve. Firm A, acting as a monopolist (by incorporating B's reaction curve in his profit-maximising computations) will produce X_A, and firm B will react by producing X_B according to its reaction curve. The sophisticated oligopolist becomes in effect the leader, while the naive rival who acts on the Cournot assumption becomes the follower. Clearly sophistication is rewarding for A because he reaches an isoprofit curve closer to his axis than if he behaved with the same naivete as his rival. The naive follower is worse off as compared with the Cournot equilibrium, since with this level of output he reaches an isoprofit curve further away from his axis.

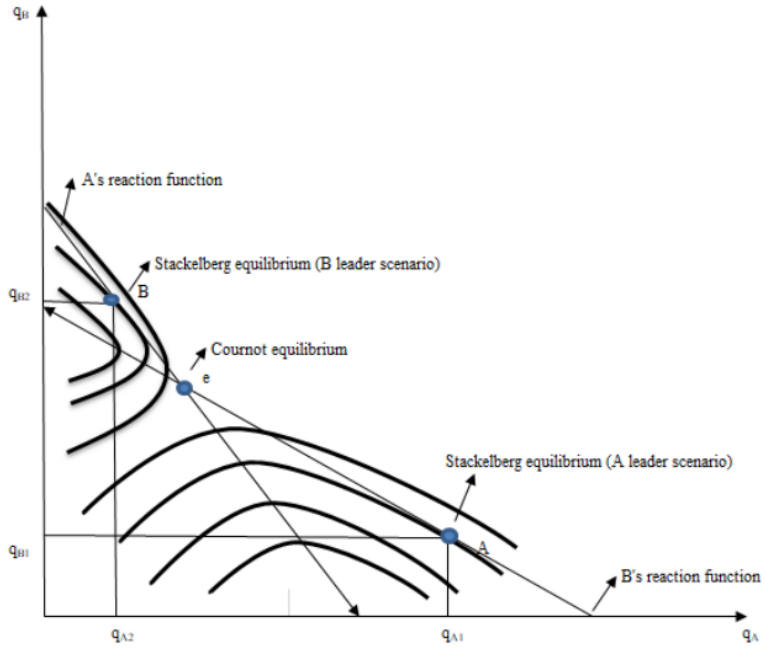


Figure 21.4

If firm B is the sophisticated oligopolist, it will choose to produce X_B, corresponding to point B on A's reaction curve, because this is the largest profit that B can achieve given his isoprofit map and A's reaction curve. Firm B will now be the leader while firm A becomes the follower. B has a higher profit and the naive firm A has a lower profit as compared with the Cournot equilibrium.

In summary, if only one firm is sophisticated, it will emerge as the leader, and a stable equilibrium will emerge, since the naive firm will act as a follower. However, if both firms are sophisticated, then both will want to act as leaders, because this action yields a greater profit to them. In this case the market situation becomes unstable. The situation is known as Stackelberg's disequilibrium and the effect will either be a price war until one of the firms surrenders and agrees to act as follower, or a collusion is reached, with both firms abandoning their naive reaction functions and moving to a point closer to (or on) the Edgeworth contract curve with both of them attaining higher profits.

Comparison of Competitive, Stackelberg, Cournot, and Collusive Equilibria

The Nash-Cournot and Nash-Stackelberg equilibrium quantities, prices, consumer surplus, and profits lie between those for the competitive (price-taking) and collusive equilibria, as Table 21.2 and Figure 21.5 show for our airline duopoly.

Table 21.2: Comparison of the Duopoly Airline Competitive, Stackelberg, Cournot, and Collusive Equilibria

	Competition	Stackelberg	Cournot	Collusion
Total output, Q (thousands)	192	144	128	96
Price, p ($)	147	195	211	243
Consumer Surplus ($ million)	18.4	10.4	8.2	4.6
Profit, π ($ million)	0	6.9	8.2	9.2

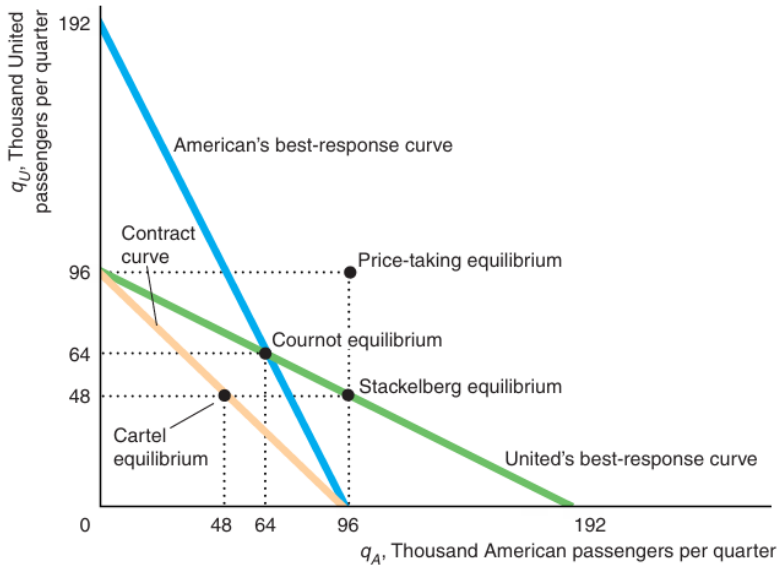


Figure 21.5 Duopoly Equilibria

If the firms were to act as price takers, they would each produce where their residual demand curve intersects their marginal cost curve, so price would equal the marginal cost of $147. The price-taking equilibrium is $q_A = q_U = 96$. If American and United were to collude, they would maximize joint profits by producing the monopoly output, $q_A + q_U = 96$. Colluding airlines could split the monopoly quantity in many ways. American could act as a monopoly and serve all the passengers, $q_A = 96$ and $q_U = 0$, and give United some of the profits. Or they could reverse roles so that United served everyone: $q_A = 0$ and $q_U = 96$. Or the two airlines could share the passengers in any combination such that the sum of the airlines' passengers equals the monopoly quantity, or, equivalently,

$$q_U = 96 - q_A$$

Figure 21.5 plots the equation, the set of possible collusive outcomes, which it labels the Contract curve. In the figure, we assume that the collusive firms split the market equally so that $q_A = q_U = 48$.

The table shows that the quantity, price, consumer surplus, and profit for the Nash-Stackelberg and Nash-Cournot equilibria lie between competition and collusion, with the Nash-Stackelberg equilibrium closer to competition and the Nash Cournot equilibrium closer to collusion. We showed that the Nash-Cournot equilibrium approaches the competitive, price taking equilibrium as the number of firms grows. Similarly, we can show that the Nash-Stackelberg equilibrium approaches the price-taking equilibrium as the number of Stackelberg followers grows. As a result, the differences between the Cournot, Stackelberg, and price-taking market structures shrink as the number of firms grows.

Multiple Choice Questions

1) What does the term "oligopoly" refer to in market structures?
    a) A market structure with many firms competing against each other.
    b) A market structure where a few firms dominate the market.
    c) A market structure with only one firm controlling the market.
    d) A market structure with no barriers to entry.

Answer: B

2) Which of the following best characterizes interdependence in oligopoly?
    a) Each firm operates completely independently of its rivals.
    b) Firms make decisions without considering the actions of competitors.
    c) Firms must consider the potential reactions of their competitors when making decisions.
    d) Firms in an oligopoly cannot set their prices freely.

Answer: C

3) What is a key feature of a collusive oligopoly?
    a) Firms compete aggressively and ignore each other's strategies.
    b) Firms reach an agreement to avoid competition and maximize joint profits.

c) Firms have complete freedom to enter and exit the market.

d) Firms produce only homogeneous products without differentiation.

Answer: B

4) Which of the following statements about barriers to entry in an oligopoly is true?

a) Barriers to entry have no impact on the market structure of oligopoly.

b) High setup costs can deter new firms from entering the market.

c) Economies of scale benefit potential entrants into an oligopoly.

d) Natural resources have no effect on the establishment of an oligopoly.

Answer: B

5) In the context of oligopoly, what is meant by the term “first-mover advantage”?

a) The advantage gained by a firm that enters the market after its competitors.

b) The benefit a firm receives by being the first to implement a new strategy, potentially leading to higher profits.

c) The strategy of following the actions of rivals to minimize risk.

d) The advantage of colluding with competitors to set prices.

Answer: B